Show that there is no other irreducible representations by finding a spanning set of $\mathrm{Cl}(V)$ with $2^{\dim V}$ elements.

(ii) Show that $\mathrm{Cl}(V)$ is semisimple if and only if $(\ ,\ )$ is nondegenerate. If $(\ ,\ )$ is degenerate, what is $\mathrm{Cl}(V)/\operatorname{Rad}(\mathrm{Cl}(V))$?

3.10. Representations of tensor products

Let A, B be algebras. Then $A \otimes B$ is also an algebra, with multiplication $(a_1 \otimes b_1)(a_2 \otimes b_2) = a_1 a_2 \otimes b_1 b_2$.

Exercise 3.10.1. Show that $\mathrm{Mat}_m(k) \otimes \mathrm{Mat}_n(k) \cong \mathrm{Mat}_{mn}(k)$.

The following theorem describes irreducible finite dimensional representations of $A \otimes B$ in terms of irreducible finite dimensional representations of A and those of B.

Theorem 3.10.2. *(i) Let V be an irreducible finite dimensional representation of A and let W be an irreducible finite dimensional representation of B. Then $V \otimes W$ is an irreducible representation of $A \otimes B$.*

(ii) Any irreducible finite dimensional representation M of $A \otimes B$ has the form (i) for unique V and W.

Remark 3.10.3. Part (ii) of the theorem typically fails for infinite dimensional representations; e.g. it fails when A is the Weyl algebra in characteristic zero. Part (i) may also fail. E.g. let $A = B = V = W = \mathbb{C}(x)$. Then (i) fails, as $A \otimes B$ is not a field.

Proof. (i) By the density theorem, the maps $A \to \operatorname{End} V$ and $B \to \operatorname{End} W$ are surjective. Therefore, the map $A \otimes B \to \operatorname{End} V \otimes \operatorname{End} W = \operatorname{End}(V \otimes W)$ is surjective. Thus, $V \otimes W$ is irreducible.

(ii) First we show the existence of V and W. Let A', B' be the images of A, B in $\operatorname{End} M$. Then A', B' are finite dimensional algebras, and M is a representation of $A' \otimes B'$, so we may assume without loss of generality that A and B are finite dimensional.

In this case, we claim that

$$\operatorname{Rad}(A \otimes B) = \operatorname{Rad}(A) \otimes B + A \otimes \operatorname{Rad}(B).$$